GOVERNMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD
II PUC EXAM – 1, MARCH 2026
SCHEME OF EVALUATION

SUBJECT: 33 - PHYSICS **MAX. MARKS: 70**

Q. No.	DESCRIPTION	MARKS
	PART – A	
	I. Pick the correct option among the four given options for **ALL** of the following questions: **15 × 1 = 15**	
1	Which one of the following charge cannot exist on a body? (a) 2e (b) 3e **(c) 3.5e** (d) −4e **ANS: (c) 3.5e**	**1**
2	The equipotential surfaces of an isolated point charge are: (a) coaxial cylindrical surfaces. (b) plane surfaces parallel to each other. **(c) concentric spherical surfaces centred at the charge.** (d) spherical surfaces but not centred on the charge. **ANS: (c) concentric spherical surfaces centred at the charge**	**1**
3	Consider the following statements about a balanced Wheatstone's bridge. Statement-I: The current through the galvanometer is zero. Statement-II: If the positions of the galvanometer and the battery are interchanged in the circuit, the current in the galvanometer will be zero. Among the above two statements: (a) Only Statement-I is true. (b) Only Statement-Il is true. (c) Both the Statements are wrong. **(d) Both the statements are true.** **ANS: (d) Both the statements are true**	**1**
4	The path traced by a charged particle moving perpendicular to a uniform magnetic field is: **(a) circle** (b) straight line (c) helix (d) ellipse **ANS: (a) circle**	**1**
5	A magnetic dipole of magnetic moment $\vec{m}$ is placed in a uniform magnetic magnetic field $\vec{B}$ such that the angle between $\vec{m}$ and $\vec{B}$ is θ. If the dipole is in stable equilibrium position, then: **(a) $\theta = 0°$** (b) $\theta = 90°$ (c) $\theta = 180°$ (d) $\theta = 45°$ **ANS: (a) $\theta = 0°$**	**1**

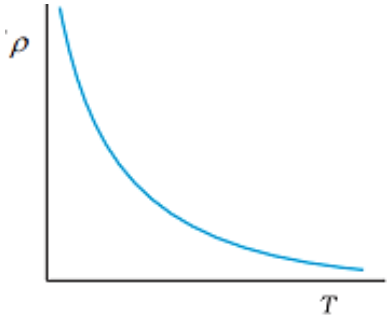
6	The working principle of an A.C. generator is: (a) mutual induction. (b) eddy currents. (c) self-induction. **(d) electromagnetic induction.** **ANS: (d) electromagnetic induction**	**1**
7	Power factor of a series *LCR* circuit is maximum when: **(a) $X_L = X_C$** (b) $X_C = 0$ (c) $X_L > X_C$ (d) $X_L < X_C$ **ANS: (a) $X_L = X_C$**	**1**
8	Displacement current is produced due to: (a) constant electric field. (b) constant magnetic field. **(c) changing electric field.** (d) changing magnetic field. **ANS: (c) changing electric field**	**1**
9	For total internal reflection of light: (a) light should be travelling from rarer medium to denser medium. **(b) light should be travelling from denser medium to rarer medium.** (c) light should be incident along the normal. (d) angle of incidence should be equal to 90°. **ANS: (b) light should be travelling from denser medium to rarer medium**	**1**
10	The angle of minimum deviation of a prism depends on: (i) refractive index of the material of the prism; (ii) refractive index of surrounding medium; (iii) refracting angle of the prism. (a) Only option (i) (b) Only option (ii) (c) Only option (iii) **(d) All (i), (ii) and (iii)** **ANS: (d) All (i), (ii) and (iii)**	**1**
11	According to Huygen's principle, speed of the secondary wavelets is: (a) twice that of the wave. (b) zero. **(c) same as that of the wave.** (d) infinite. **ANS: (c) same as that of the wave**	**1**
12	An α-particle, a proton, an electron and a neutron are moving with the same velocity. Then the particle having longest de Broglie wavelength is: (a) proton. **(b) electron.** (c) neutron. (d) α-particle. **ANS: (b) electron**	**1**
13	Let *K* be the kinetic energy, *U* be the potential energy and *E* be the total energy of an electron revolving around the nucleus in a hydrogen atom, then which of the following is correct? (a) $K > 0$, $U > 0$, $E > 0$ **(b) $K > 0$, $U < 0$, $E < 0$** (c) $K > 0$, $U > 0$, $E < 0$ (d) $K < 0$, $U < 0$, $E < 0$ **ANS: (b) $K > 0$, $U < 0$, $E < 0$**	**1**

14	An example for isobars is: (a) $^{2}_{1}$H and $^{3}_{1}$H (b) $^{2}_{1}$H and $^{4}_{2}$He (c) $^{198}_{80}$Hg and $^{197}_{79}$Au **(d) $^{3}_{1}H$ and $^{3}_{2}He$** **ANS: (d) $^{3}_{1}$H and $^{3}_{2}$He**	**1**
15	Which of the following pairs are elemental semiconductors? (a) Silicon and aluminium **(b) Silicon and germanium** (c) Germanium and cadmium (d) Aluminium and cadmium **ANS: (b) Silicon and germanium**	**1**
II. Fill in the blanks by choosing appropriate answer from the given options for all the following questions: **5 × 1 = 5** **(*photons, diffraction, polarity, monopoles, greater than unity, less than unity*)**		
16	According to Gauss's law in magnetism, magnetic ***monopoles*** are not known to exist.	**1**
17	Lenz's law gives the ***polarity*** of induced emf.	**1**
18	For a step-down transformer, the ratio of primary current to secondary current is ***less than unity***.	**1**
19	The bending of light around the corners and entering into the geometric shadow region is called ***diffraction***.	**1**
20	In interaction with matter, light behaves as if it is made up of packets of energy called ***photons***.	**1**
PART - B		
III. Answer **ANY FIVE** of the following questions: **5 × 2 = 10**		
21	**Write any two properties of electric field lines.** (i) Electric field lines start from positive charge and end on negative charge. (ii) In charge free region, electric field lines are continuous curves without any breaks. (iii) Electrostatic field lines do not form closed loops. (iv) Electric field lines do not intersect each other. (v) Tangent to the electric field line at any point gives the direction of electric field at that point. (vi) The density of field lines at any point is a measure of the electric field at that point. **(Any two; Any other correct property has to be considered)**	**1 + 1**
22	**What is electrostatic shielding? Mention one use of it.** The electric field inside the cavity of a conductor is zero (OR Electrostatic field inside a conductor is zero). This is called electrostatic shielding. Sensitive electric instruments are placed inside metal enclosures to preserve them from outside electric influence.	**1** **1**

23

How does the resistivity of a semiconductor vary if its temperature is increased? Also show the variation graphically.

As the temperature increases, the resistivity of a semiconductor decreases. **1**



1

24

A thin long straight wire carries a current of 5 A. Calculate the magnitude of the magnetic field at a distance of 0.25 m from the wire. ($\mu_o = 4\pi \times 10^{-7}$ Hm^{-1})

The magnitude of the magnetic field at a point due to a straight wire,

$$B = \frac{\mu_o I}{2\pi r}$$ **1**

$$\Rightarrow B = \frac{(4\pi \times 10^{-7})(5)}{(2\pi)(0.25)} = 4 \times 10^{-6}\ \text{T}$$ **1**

25

Write the expression for the motional emf induced in a straight conductor moving perpendicular to a uniform magnetic field. Explain the terms.

Motional emf,

$$\varepsilon = Blv$$ **1**

where B = magnetic field; l = length of the rod; v = speed of the rod **1**

26

Name the electromagnetic waves used in:

i) the radar systems of aircrafts. ii) the laser assisted eye surgery (LASIK).

Radar system of aircrafts → Microwaves **1**

LASIK surgery → Ultraviolet rays **1**

27

Define impact parameter. What is the angle of scattering of an α-particle for zero impact parameter?

The impact parameter is the perpendicular distance of the initial velocity vector of the α-particle from the centre of the target nucleus. **1**

Angle of scattering is 180°. **1**

28

List out two differences between p-type and n-type semiconductors.

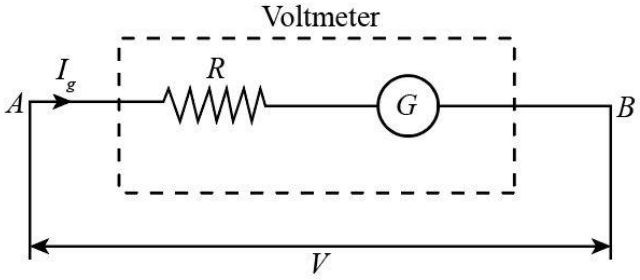
***p* – type semiconductors**	***n* – type semiconductors**
1. They are obtained by doping pure semiconductors with trivalent impurity.	1. They are obtained by doping pure semiconductors with pentavalent impurity.
2. Holes are the majority charge carriers.	2. Electrons are the majority charge carriers.
3. Electrons are the minority charge carriers.	3. Holes are the minority charge carriers.

(Any two; Any other correct difference must be considered)

1 + 1

PART - C

IV. Answer ANY FIVE of the following questions: **5 × 3 = 15**

29	**Give Coulomb's law in vector form and explain the terms. Define SI unit of charge.** Electrostatic force on charge q_2 by charge q_1 is: $$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r_{21}^2}\hat{r}_{21}$$ where q_1, q_2 are charges, r_{21} is the distance between the charges, $\hat{r}_{21}$ is the unit vector in the direction from q_1 towards q_2. One coulomb is the charge on two identical point charges which are separated by a distance of 1 m in vacuum experience an electric force of repulsion of 9×10^9 N. (The definition: ***One coulomb is the charge flowing through a wire in 1 s if the current is 1 A*** should also be considered)	**1** **1** **1**
30	**What is a capacitor? Mention any two factors on which capacitance of a parallel plate capacitor depends.** A capacitor is a system of two conductors separated by an insulator. (**OR** Capacitor is a device that is used to store charges.) Capacitance of a capacitor depends on: (i) Area of the plates; (ii) Distance between the plates; (iii) Medium between the plates (or dielectric constant of the medium between the plates). **(Any two factors)**	**1** **1** + **1**
31	**With the help of a circuit diagram, explain how to convert a galvanometer into a voltmeter.** A galvanometer can be converted into a voltmeter by connecting a suitable high resistance in series with it. The circuit diagram is as shown below:  The value of the series resistance is given by: $$R = \frac{V}{I_g} - G$$ where V = range of the voltmeter; I_g = maximum current through the galvanometer and G = resistance of the galvanometer. (**Other symbols should also be considered**)	**1** **1** **1**

32 **Differentiate between diamagnetic and ferromagnetic materials.**

Diamagnetic Materials	Ferromagnetic Materials
(i) These are the materials that are weakly repelled by a magnetic field.	(i) These are the materials that are strongly attracted by a magnetic field.
(ii) Magnetic susceptibility values for them is negative. (OR $\chi_m < 0$)	(ii) Magnetic susceptibility values for them is positive. (OR $\chi_m \gg 0$)
(iii) Relative permeability values for them have ranges $0 \leq \mu_r < 1$.	(iii) Relative permeability values for them have ranges $\mu_r > 1$.
(iv) There is no formation of domains.	(iv) There is formation of domains.
(v) Atoms of these materials do not have net magnetic moment.	(v) Atoms of these materials have net magnetic moment.
(vi) When placed in an external magnetic field, the magnetic field inside these materials is very much less than the value of the field outside.	(vi) When placed in an external magnetic field, the magnetic field inside these materials is very much greater than the value of the field outside.

(Any three; Any other correct difference has to be considered)

1 + 1 + 1

33 **Obtain the expression for energy stored in an inductor.**

During setting-up a current of I in a coil of self-inductance L, the instantaneous power is:

$$p = |\varepsilon| i = Li\frac{di}{dt}$$ **1**

Therefore, the magnetic energy stored in the coil (which is equal to energy spent in the process) is:

$$U = \int p\, dt = \int_0^I Li\frac{di}{dt}$$ **1**

$$\Rightarrow U = \frac{1}{2}LI^2$$ **1**

34 **A refracting telescope has an objective lens of focal length 144 cm and the length of the tube is 150 cm. Calculate the magnification due to the telescope.**

Given: $f_o = 144$ cm; $L = 150$ cm

We know that:

$$L = f_o + f_e$$

$$\Rightarrow f_e = L - f_o = 150 - 144 = 6 \text{ cm.}$$ **1**

The magnification is:

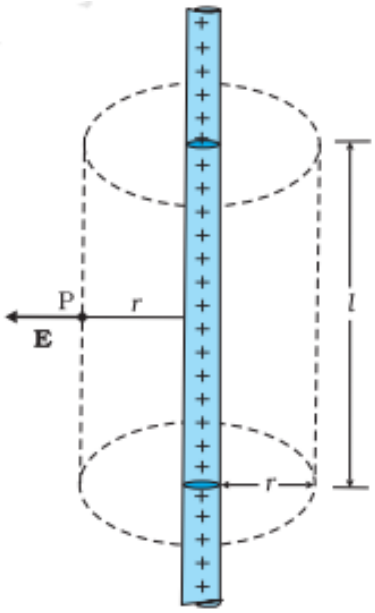
$$m = \frac{f_o}{f_e}$$ **1**

$$\Rightarrow m = \frac{144}{6} = 24$$ **1**

35	**Write the experimental observations of photoelectric effect.** The experimental observations of photoelectric effect are: (i) Photoelectric effect is an instantaneous process. (ii) For a given frequency of incident light, photoelectric current is directly proportional to the intensity of incident light. (iii) For any metal surface there exists a minimum frequency of incident light (or threshold frequency) below which there is no photoelectric emission. (iv) Above threshold frequency, the maximum kinetic energy (or stopping potential) of photoelectrons is directly proportional to the frequency of incident light. (v) The maximum kinetic energy of photoelectrons is independent of the intensity of the incident light when the frequency is kept constant. (vi) The photoelectric current is independent of the frequency of incident radiation. **(Any three)**	**1** + **1** + **1**
36	**List three conclusions drawn from observations of binding energy per nucleon versus mass number curve.** The conclusions that can be drawn from the observations of binding energy curve are: (i) Nuclear force is attractive and strong force. (ii) Nuclear force is short ranged. (iii) Nuclear force shows the property of saturation. (iv) Light nuclei undergo nuclear fusion to release energy. (v) Heavy nuclei undergo nuclear fission to release energy. **(Any three)**	**1** + **1** + **1**

PART - D

V. Answer ANY THREE of the following questions: **3 × 5 = 15**

37	**Arrive at the expression for the electric field at a point due to an infinitely long uniformly charged straight wire using Gauss's law.** Consider an infinitely long uniformly charged straight wire with uniform linear charge density λ. Let us calculate the electric field at point P at a distance r from this wire. We will consider a cylindrical gaussian surface of radius r and length l as shown in the figure.  	**1**

By definition, electric flux through this cylinder is given by:

$$\phi_E = \int \vec{E} \cdot d\vec{A} = E \times 2\pi rl$$ **1**

By Gauss's law in electrostatics, flux though the closed surface is:

$$\phi_E = \frac{q_{enc}}{\varepsilon_o}$$ **1**

But enclosed charge, $q_{enc} = \lambda l$.

$$\Rightarrow \phi_E = \frac{\lambda l}{\varepsilon_o}$$ **1**

Comparing the two equations, we get:

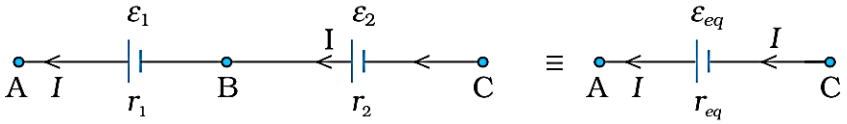
$$E \times 2\pi rl = \frac{\lambda l}{\varepsilon_o}$$

$$\Rightarrow E = \frac{\lambda}{2\pi\varepsilon_o r}$$ **1**

38

Obtain the expression for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in series.

Consider two cells ε_1, r_1 and ε_2, r_2 connected in series as shown in the figure. Let the equivalent cell be ε_{eq}, r_{eq}. Let the terminal potential differences for the first cell be V_1, for the cell be V_2 and for combination be V.



1

For the first cell, $V_1 = \varepsilon_1 - I\, r_1$ and for the second cell, $V_2 = \varepsilon_2 - I\, r_2$. **1**

As the cells are in series, $V = V_1 + V_2$.

$$\Rightarrow V = (\varepsilon_1 - I\, r_1) + (\varepsilon_2 - I\, r_2) = (\varepsilon_1 + \varepsilon_2) - I(r_1 + r_2).$$ **1**

For the equivalent cell, $V = \varepsilon_{eq} - I\, r_{eq}$. **1**

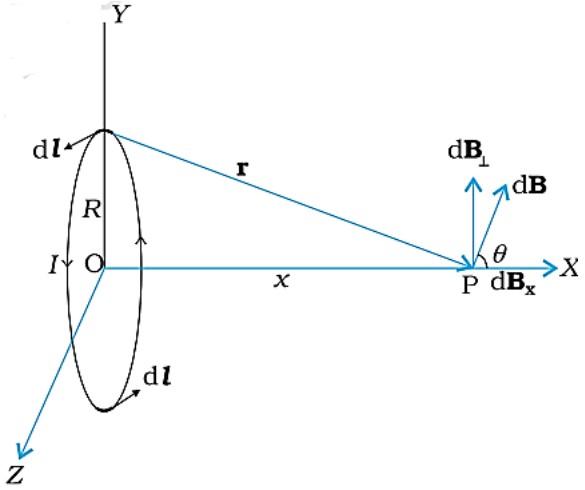
By comparing the last two equations, we get:

$$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2 \quad \text{and} \quad r_{eq} = r_1 + r_2$$ **1**

39

Derive the expression for the magnitude of the magnetic field at a point on the axis of a circular loop carrying current.

Consider a circular loop of radius R carrying a current I placed in YZ plane. Let us calculate the magnetic field at a point P along the axis of the loop. The distance of the point P from the centre of the circular loop is x.



1

The magnitude of the magnetic field at P due to a small element of the loop shown is:

$$dB = \frac{\mu_o}{4\pi}\frac{I\left|d\vec{l}\times\vec{r}\right|}{r^3} = \frac{\mu_o}{4\pi}\frac{I\,dl}{r^2}$$ **1**

When we add $d\vec{B}$ to get the total magnetic field at P, the vertical components of $d\vec{B}$ cancel and horizontal components (along x - axis) get added up. **1**

The x - component of $d\vec{B}$ is:

$$dB_x = \frac{\mu_o}{4\pi}\frac{I\,dl}{r^2}\cos\theta$$

The total magnetic field at P is:

$$B = \sum dB_x = \sum \frac{\mu_o}{4\pi}\frac{I\,dl}{r^2}\cos\theta$$ **1**

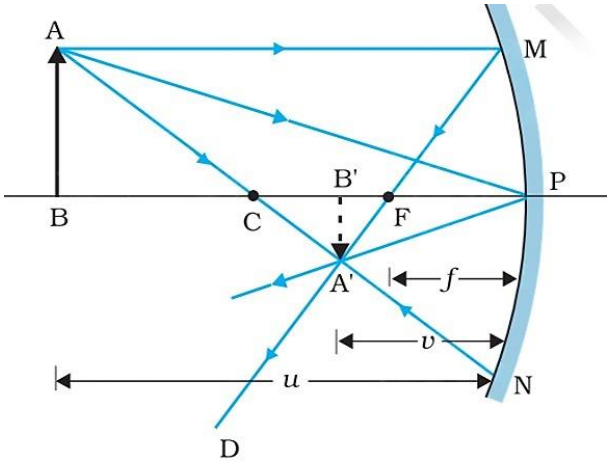
But $\sum dl = 2\pi R, \cos\theta = \frac{R}{r}$ and $r = (R^2 + x^2)^{1/2}$. Substituting these into the above equation, we get:

$$B = \frac{\mu_o I R^2}{2(R^2 + x^2)^{3/2}}$$ **1**

40

Derive mirror equation $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ for a concave mirror.

Consider an object AB placed on the principal axis of a concave mirror as shown in the figure. The ray diagram for image formation for paraxial rays is as shown in the figure. A'B' is the image of the object as shown. F is the focal point of the mirror. u is the object distance and v is the image distance.



1

From tringles triangles A'B'F and MPF,

$$\frac{\mathrm{B'A'}}{\mathrm{PM}} = \frac{\mathrm{B'F}}{\mathrm{FP}} \Rightarrow \frac{\mathrm{B'A'}}{\mathrm{AB}} = \frac{\mathrm{B'F}}{\mathrm{FP}} \Rightarrow \frac{\mathrm{B'A'}}{\mathrm{BA}} = \frac{\mathrm{B'F}}{\mathrm{FP}}$$ **1**

Also from triangles A'B'P and ABP,

$$\frac{\mathrm{B'A'}}{\mathrm{BA}} = \frac{\mathrm{B'P}}{\mathrm{BP}}$$ **1**

$$\Rightarrow \frac{\mathrm{B'P} - \mathrm{FP}}{\mathrm{FP}} = \frac{\mathrm{B'P}}{\mathrm{BP}}$$ **1**

From figure, B'P = – v, BP = – u and FP = – f.

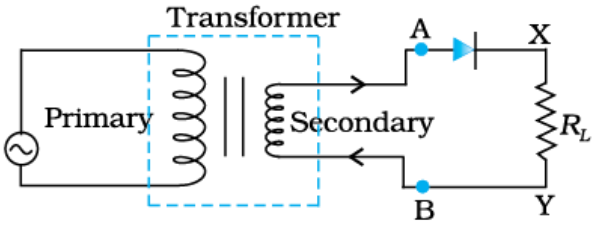
Substituting these and rearranging, we get:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$ **1**

41 **What is half wave rectifier? Describe with a circuit diagram, the working of a semiconductor diode as a half wave rectifier Draw input and output waveforms.**

A half-wave rectifier is a device (circuit) that rectifies only one half cycle of AC. **1**

Construction: The circuit diagram for a half-wave rectifier using a diode and a transformer is as shown below.

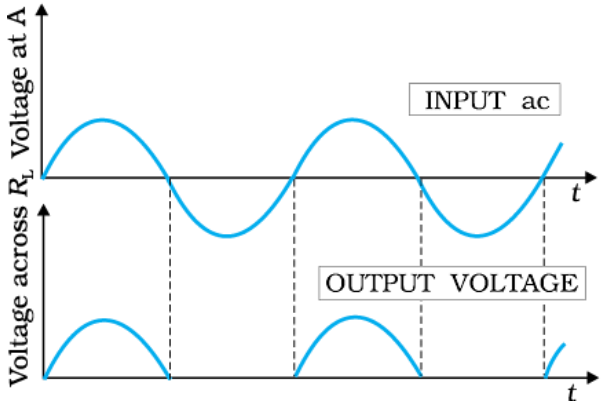


1

Working: During positive half cycle of AC, A is positive and B is negative. The diode becomes forward biased and it conducts. The output appears across the resistance R_L. **1**

During negative half cycle of AC, A becomes negative and B becomes positive. The diode is reverse biased, and it does not conduct. No output appears in this case. **1**

The input and output waveforms are as shown in the figure.



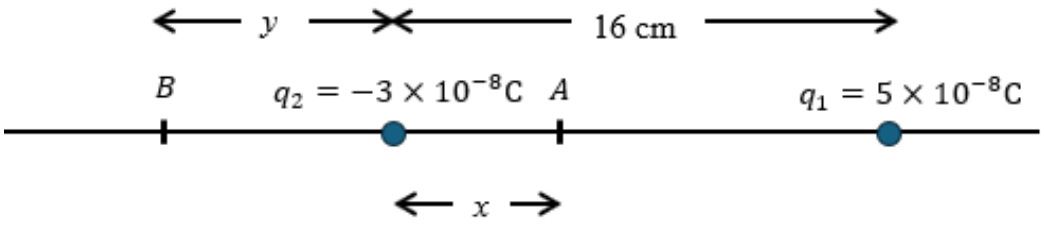
1

VI. Answer ANY TWO of the following questions: **2 × 5 = 10**

42 **Two charges 5×10^{-8} C and -3×10^{-8} C are located 16 cm apart in vacuum. Find the positions along the line passing through the two charges at which the electric potential is zero.**

Electric potential due to a point charge, **1**

$$V = \frac{1}{4\pi\varepsilon_o}\frac{Q}{r}$$



	Let the potential be zero at points A and B around the charge of smaller magnitude. For $V_A = 0$, $$\frac{1}{4\pi\varepsilon_o}\frac{q_1}{16-x}+\frac{1}{4\pi\varepsilon_o}\frac{q_2}{x}=0$$	**1**
	$$\Rightarrow \frac{5\times10^{-8}}{16-x}+\frac{-3\times10^{-8}}{x}=0 \Rightarrow \frac{5}{16-x}-\frac{3}{x}=0$$ $$\Rightarrow x = 6 \text{ cm}$$	**1**
	For $V_B = 0$, $$\frac{1}{4\pi\varepsilon_o}\frac{q_1}{16+y}+\frac{1}{4\pi\varepsilon_o}\frac{q_2}{y}=0$$	**1**
	$$\Rightarrow \frac{5\times10^{-8}}{16+y}+\frac{-3\times10^{-8}}{y}=0 \Rightarrow \frac{5}{16+y}-\frac{3}{y}=0$$ $$\Rightarrow y = 24 \text{ cm}$$	**1**
	(Distances from 5×10^{-8} C charge are 10 cm and 40 cm to be considered) **(Any other correct alternative should be considered)**	
43	**The power dissipated across a wire of length 0.5 m and area of cross-section 0.2×10^{-6} m² is 10 W when a steady current flows through it. Calculate the following quantities:** **(Given: resistivity of the material of the wire is 1×10^{-6} Ω m)** **i) Resistance of the wire.** **ii) The current flowing through the wire:** **iii) The current density for the wire.** Given: $l = 0.5$ m, $A = 0.2 \times 10^{-6}$ m², $P = 10$ W, $\rho = 1 \times 10^{-6}$ Ω m (i) Resistance of the wire, $$R = \frac{\rho l}{A}$$	**1**
	$$\Rightarrow R = \frac{(1\times10^{-6})\times(0.5)}{0.2\times10^{-6}} = 2.5\ \Omega$$	**1**
	(ii) Current through the wire, $$P = I^2R$$	**1**
	$$\Rightarrow I = \sqrt{\frac{P}{R}} = \sqrt{\frac{10}{2.5}} = 2 \text{ A}$$	**1**
	(iii) Current density is: $$J = \frac{I}{A} = \frac{2}{0.2\times10^{-6}} = 10\times10^6 \text{ Am}^{-2} \quad (OR\ 1\times10^7 \text{ Am}^{-2})$$	**1**
	(Any other correct alternative should be considered)	
44	**An inductor and a resistor are connected in series with 200 V, 50 Hz a.c. source. The current in the circuit is 2 A and voltage leads the current by π/3. Calculate the inductance of the inductor.**	

	Given: $\phi = -\pi/3$, $I = 2$ A, $V = 200$ V, $f = 50$ Hz. Impedance of the circuit, $$Z = \frac{V}{I} = \frac{200}{2} = 100\ \Omega$$ $$\sin\phi = \frac{X_L}{Z}$$ $$\Rightarrow \sin\frac{\pi}{3} = \frac{X_L}{Z} \Rightarrow X_L = \frac{\sqrt{3}}{2}Z$$ $$\Rightarrow X_L = 86.6\ \Omega$$ $$X_L = 2\pi f L$$ $$L = \frac{X_L}{2\pi f} = \frac{86.6}{2\pi \times 50} = 0.276\ \text{H} \quad \textbf{OR} \quad 0.28\ \text{H}$$ **(Any other correct alternative should be considered)**	**1** **1** **1** **1** **1**
45	**A beam of light consisting of two wavelengths 500 nm and 600 nm is used to obtain interference fringes in Young's double slit experiment. Distance between the slits is 1 mm and the screen is placed at a distance of 1.2 m from the slits.** **i) Find the least distance between the central maximum and the point where the bright fringes due to both the wavelengths coincide.** **ii) Find the distance of the third dark fringe from the central bright fringe for the first wavelength.** Given: $\lambda_1 = 500$ nm, $\lambda_2 = 600$ nm, $d = 1$mm, $D = 1.2$ m (i) Fringes coinciding $\Rightarrow$ $$\frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$$ $$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{600}{500} = \frac{6}{5}$$ Minimum distance $\Rightarrow n_1 = 6; n_2 = 5.$ Required distance, $$x = \frac{n_1\lambda_1 D}{d}$$ $$\Rightarrow x = \frac{(6) \times (500 \times 10^{-9}) \times (1.2)}{(1 \times 10^{-3})} = 3.6 \times 10^{-3}\text{m}$$ (ii) The distance of the third dark fringe from the central bright fringe, $$x' = \frac{(2n_1 - 1)\lambda_1 D}{2d}$$ $$\Rightarrow x' = \frac{(2 \times 3 - 1) \times (500 \times 10^{-9}) \times (1.2)}{2 \times (1 \times 10^{-3})} = 1.5 \times 10^{-3}\text{m}$$ (For part (ii), using the formula $x' = \left(n + \frac{1}{2}\right)\frac{\lambda D}{d}$ with $n = 2$ to get the correct answer should be considered) **(Any other correct alternative should be considered)**	**1** **1** **1** **1** **1**